CONFORMAL TRANSFER AND CURVATURE-RESPONSE CLOSURE FOR AXISYMMETRIC EULER FLOWS IN PERIODIC VARIABLE-RADIUS PIPES

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the regularity and finite-time singularity problem for the three-dimensional incompressible Euler equations, we investigate the transfer of the Chen–Hou axisymmetric boundary blowup framework from a straight periodic cylinder to periodic variable-radius Euclidean pipes $P_\kappa = \{(R, \vartheta, Z) : 0 \le R < 1 + \kappa(1 - \cos Z),\ Z \in \mathbb{R}/2\pi\mathbb{Z}\}$. We first construct an explicit family of stream functions with vanishing origin jets that refutes the direct Cartesian weighted Biot–Savart inverse estimate via a logarithmic divergence. We then introduce an adapted conformal coordinate chart that eliminates the anisotropic principal part and prove closed weighted inverse, transport, and curvature-response bounds uniformly for $0 < |\kappa| \le 1/10$.

1. Main Mathematical Results

Theorem 1.1 (Cartesian logarithmic obstruction and conformal response closure)**.** *On the periodic variable-radius pipe P_κ ($0 < |\kappa| \le 1/10$):*

(1) *The uncorrected Cartesian weighted inverse operator exhibits a linear-in-mode amplification on zero-flux stream functions with vanishing origin jets, refuting the naive cylindrical transfer inequality;*

(2) *In the adapted conformal chart (ρ, η), the transformed stream-vorticity inverse, singular weight transport, and curvature-coupling remainder satisfy closed quantitative energy and holder-weighted bounds across all separate angular modes.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 40 in the domain of partial differential equations and fluid dynamics adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Logarithmic counterexample to Cartesian weighted inverse bounds and exact conformal transport/curvature-response closure on periodic cosine-wall pipes 0 < |kappa| <= 1/10.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted 3D incompressible Euler global regularity in R^3 without boundary

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 40 (partial differential equations and fluid dynamics), archived in the accompanying repository package `results/40/`.

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